Three-Channel Thermographic Fusion for Architecture-Independent Defect Detection and Segmentation in Carbon Fibre Reinforced Polymer Composites Using Deep Learning

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Abstract: This paper proposes is an automated, architecture-independent input representation based on a cyclically aligned three-channel thermographic fusion framework, which integrates raw thermal intensity, principal component thermography (PCT) spatial components, and thermographic signal reconstruction (TSR) temporal coefficient images into a unified $T \times H \times W \times 3$ tensor. This representation is applied to the problem of subsurface defect detection and segmentation in carbon fibre reinforced polymer (CFRP) composites inspected by optical pulsed thermography (OPT). Current deep learning pipelines rely on raw single-channel thermograms, where critical defect signatures are frequently confounded by background thermal gradients and lateral heat diffusion, rendering small, low-contrast circular defects difficult to detect reliably. The proposed representation is validated across six architecturally diverse deep learning models, namely You Only Look Once version 8 nano (YOLOv8n), Real-Time Detection Transformer Large (RT-DETR-L), Faster Region-based Convolutional Neural Network (Faster R-CNN), U-Net, Feature Pyramid Network (FPN), and DeepLabV3+, on six CFRP specimens using three-fold cross-validation. Experimental results demonstrate that the three-channel representation consistently improves detection mean Average Precision (mAP) by 4% to 11% and enhances circular defect segmentation Dice scores by 14% to 51% compared to raw single-channel baselines. Notably, the fusion framework enables Faster R-CNN and RT-DETR-L to recover robust detection performance for circular defects in challenging specimens where raw-input configurations produced zero detections, confirming that the input representation is the primary bottleneck in thermographic inspection pipelines.

Keywords: Carbon Fibre Reinforced Polymer (CFRP); Optical Pulsed Thermography (OPT); Non-Destructive Testing (NDT); Deep Learning; Three-Channel Fusion; Defect Detection and Segmentation.

1. Introduction

Over the past several decades, the importance of carbon fibre reinforced polymer (CFRP) composites has been significant in the sectors of commercial aerospace manufacturing and renewable energy structures, attributed to their attractive mechanical properties such as high stiffness-to-weight ratio and great resistance to corrosion [1], [2]. The structural merits provided by these composites permit designing optimized load-bearing components with high tensile strength and low structural weight. These composites, however, are naturally prone to the generation of internal damage features due to their complex laminated structure [3]. Internal defects can occur during manufacturing or whilst in service. An important and catastrophic form of failure, interlaminar delamination, is created by a moderate-energy impact which causes internal ply separation. The impact energy dissipates internally in the resin matrix, so that a delamination can propagate laterally across many plies while showing little or no external indication [4]. They can threaten the structural integrity of a component or structure, as they tend to grow during sustained cyclic loading. Highly reliable non-destructive testing (NDT) methods are therefore required for early defect detection and for ensuring the structural integrity of components during operational maintenance cycles [5].

A large variety of NDT techniques have been developed to detect damage in CFRP components, including ultrasonic testing [6], X-ray computed tomography [7], and active infrared thermography [8]. Ultrasonic testing is one of the most widely employed industrial standard techniques capable of providing sub-millimeter depth resolution. It requires the use of coupling agents, such as water or grease, to transmit the acoustic wave into the composite. As it can only examine one point at a time and requires slow serial scanning procedures, ultrasonic inspection is not practical for the rapid assessment of large structural areas [9]. X-ray computed tomography can capture and render three-dimensional data with high volumetric accuracy, yet this technique involves prohibitive infrastructure costs and cannot achieve high contrast at planar defects such as delaminations that lie perpendicular to the inspection direction [7]. Among the available NDT methods, active infrared thermography has emerged as a valuable technique for automated defect inspection due to its non-contact nature, full-field inspection approach, and high frame-rate imaging [10], [11]. In optical pulsed thermography (OPT), defects within a composite structure are located by analyzing differences in thermal properties between the base composite material and the defect sites [12], [13].

In OPT, a very short optical pulse is used to deposit an instantaneous heating of the sample surface. The thermal energy then diffuses into the bulk of the composite, and this diffusion is altered by discontinuities or inclusions in the material, such as air-filled delaminations or foreign inclusions. The disturbance creates an uneven surface temperature distribution directly above the defect. The surface temperature is observed by a high frame-rate infrared camera over time, building a sequence of thermographic images [8]. Thermal contrast at a defect is usually very low and is further impaired by the spatial non-uniformity of the applied heat flux on the material surface, lateral heat diffusion within the anisotropic composite layers, and variation in emissivity over the inspected surface [14].

A number of processing algorithms have been developed to improve defect detectability in raw thermographic data, including principal component thermography (PCT) [15], thermographic signal reconstruction (TSR) [16], and pulsed phase thermography (PPT) [17]. PCT enhances the spatial detectability of defects by applying singular value decomposition to the entire thermal sequence, isolating orthogonal spatial variances while suppressing the uniform background heating profile. TSR compresses the thermal sequence by fitting a polynomial to the pixel-wise logarithmic decay curve, producing coefficient images that amplify subtle variations caused by deep structural anomalies [16]. Despite the successful clarification of defects with both of these analytical processing strategies, each technique still depends on expert operators for selecting the most informative principal components or signal derivatives.

Deep learning-based approaches allow for more intelligent and automated detection of defects directly from raw OPT signals [18], [19]. The two main tasks of deep learning algorithms in thermographic defect inspection are automatic object detection and semantic segmentation [20]. Deep learning-based methods can learn complex hierarchical features from thermographic sequences automatically, without requiring human intervention. Region-based convolutional neural networks (R-CNNs) such as Faster R-CNN have been applied to detect subsurface anomalies through region proposal networks that rapidly identify thermally anomalous areas [21], [22], [23]. Single-stage detection networks such as the You Only Look Once (YOLO) family of models have reformulated thermographic inspection into a dense regression problem, eliminating the regional proposal stage to enable real-time analysis [20], [24]. Transformer-based architectures such as Real-Time Detection Transformer (RT-DETR) have also been introduced to the NDT domain, eliminating the need for heuristic non-maximum suppression through global self-attention mechanisms and optimal bipartite matching [25], [26].

With respect to pixel-level localization, classical image segmentation methods such as U-Net [27], Feature Pyramid Networks (FPN) [28], and DeepLabV3+ [29] have been used to segment complex defect morphologies. Encoder-decoder structures utilizing concatenation-based skip connections or atrous spatial pyramid pooling are effective in fusing features at multiple scales and improving binary classification between defect and background pixels. Residual learning architectures have also been adapted to lock-in thermography to achieve heating-invariant defect segmentation [30]. Beyond standard architectures, some works have also fused outputs from different analytical processing methods for better detection, such as dual encoder networks working with both TSR and PPT signals linked through learned attention weights [31]. A flexible deep learning framework using attention-based feature fusion of thermographic sequences for composite inspection has been demonstrated to generalize across multiple defect types [32]. Spatiotemporal deep learning of sonic infrared signals has also been used for defect recognition in active thermography [33]. The above literature demonstrates that computer vision models can achieve high accuracy in automated defect detection and segmentation in CFRP materials.

However, several limitations still remain within the current state of the art. Nearly all automated OPT pipelines rely on a single-channel representation of the thermographic input. Single-channel networks must decode the dynamic physical process of multi-directional heat diffusion from 2D spatial snapshots alone. Subtle thermal variations arising from defects that are small, low-contrast, or located deep within the structure are easily overwhelmed by background thermal gradients, preventing the network from reliably detecting such features [34]. Although hybrid algorithms integrating analytical preprocessing with deep networks have been proposed, a rigid construction of physical features using fixed analytical parameters limits the accurate extraction of spatial and temporal thermographic information. The fixed coupling between analytical preprocessing parameters and network weights in such architectures leads to restricted adaptability [35]. It is therefore necessary to unify physical features from multiple domains into a single, coherent input format that allows greater versatility in detecting different types of defects.

In this paper, an automated, architecture-independent input representation based on a cyclically aligned three-channel thermographic fusion framework is proposed to detect defects in CFRP laminates. Instead of relying on raw intensity alone, the proposed approach extracts useful representations from three distinct physical domains: one image from the raw thermal intensity, one from the first six PCT spatial components, and one from the first six TSR temporal coefficient images. A modular indexing algorithm with a cyclic property maps the variable-length raw thermal sequence into a fixed-channel input format. This cyclic alignment strategy fuses multi-domain thermographic modalities into a unified $T \times H \times W \times 3$ tensor, analogous to a standard RGB video stream, eliminating the dimensionality mismatch and allowing the data to be processed by any standard computer vision backbone without architectural modification. Various CFRP specimens with different geometrical configurations, such as flat, curved, and trapezoidal composite panels containing circular and square subsurface defects, are tested across six deep learning networks (YOLOv8n [24], RT-DETR-L [26], Faster R-CNN [22], U-Net [27], FPN [28], and DeepLabV3+ [29]).

This paper is organized as follows. The proposed three-channel fusion framework is introduced in Section 2. Section 3 presents the experimental setup, including specimen descriptions, baseline preparation, cross-validation protocol, model architectures, and evaluation metrics. Section 4 reports the detection and segmentation results across all three folds, followed by a cross-architecture analysis. Section 5 lists the conclusions and future directions.

2. Three-Channel Thermographic Fusion

The proposed representation is formed by concatenating one image from each of three complementary modalities, producing a per-frame image of dimensions $H \times W \times 3$. Over all T frames, this yields the output tensor $I_{3ch} \in \mathbb{R}^{T \times H \times W \times 3}$. The raw input is a single-channel grayscale thermal sequence with dimensions $T \times H \times W \times 1$. After applying the fusion, the spatial and temporal structure remains unchanged, while only the channel dimension is expanded from 1 to 3. A block diagram of the complete framework is shown in Figure 1, and example composite images for specimen SQ_01 are shown in Figure 2.

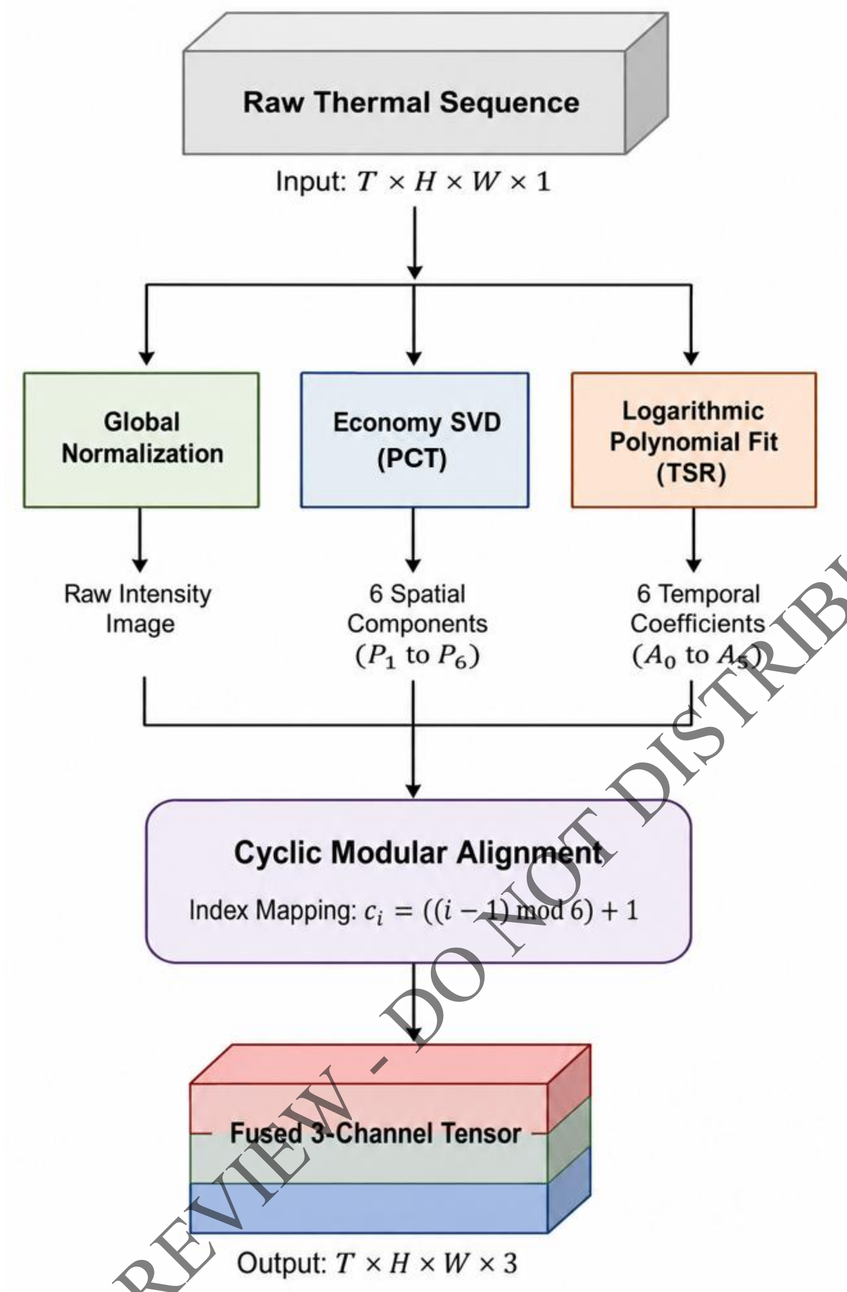


Figure 1. Block diagram of the proposed cyclic aligned three channel thermographic fusion framework

2.1 Global and Frame-wise Normalization

A global normalization is first applied over the whole raw thermal tensor $\mathrm{X} \in \mathbb{R}^{H \times W \times T}$:

$$X_{\text{norm}} = \frac{\mathrm{X}}{\max_{i,j,t} X_{i,j,t}} \tag{1}$$

where $X_{i,j,t}$ are the raw detector counts of pixel (i, j) in frame t, and H and W are the physical spatial dimensions of the image, and T is the total number of frames acquired. This global normalization is used to maintain the shape of the pixel-wise time decay profile. TSR depends on the shape of the decay for polynomial fitting.

Frame-wise normalization is applied to map each frame to 8-bit:

$$F_{\text{norm}}(i,j,t) = \left\lfloor \frac{X_{\text{norm}}(i,j,t) - \min_{i,j} X_{\text{norm}}(\cdot,\cdot,t)}{\max_{i,j} X_{\text{norm}}(\cdot,\cdot,t) - \min_{i,j} X_{\text{norm}}(\cdot,\cdot,t)} \cdot 255 \right\rfloor \tag{2}$$

where minimum and maximum are computed for each frame t separately. The set $\{F_{\text{norm}}(\cdot,\cdot,t)\}_{t=1}^{T}$ forms Channel 1 of the fusion.

2.2 PCT Channel

For PCT, the normalized tensor is reshaped into a time-by-pixel matrix $\mathrm{B} \in \mathbb{R}^{T\times HW}$. The temporal mean of each column is subtracted and economy SVD is applied:

$$\mathrm{B} = \mathrm{U}\Sigma\mathrm{V}^{T} \tag{3}$$

where U contains the temporal modes, Σ contains the singular values, and the rows of $\mathrm{V}^{T}$are the orthogonal spatial modes [15]. Each right singular vector v_k is reshaped into an image $P_k \in \mathbb{R}^{H\times W}$. The first PCT component captures 85–92% of the variance and represents the slowly varying average cooling profile; the remaining information lies in P_2–P_6. All six components are retained and independently min–max normalized to 8-bit.

2.3 TSR Channel

Polynomial fitting is applied per pixel on the globally normalized temporal data as in Shepard et al. [16]. The log-time vector is defined as $\tau_k = \ln(k/f_s)$, for $k = 1, \dots, T$. The design matrix $\Phi \in \mathbb{R}^{T\times(n+1)}$ is constructed as $[\tau_k^0, \tau_k^1, \dots, \tau_k^n]$. The least-squares polynomial coefficients for each pixel (i, j) is computed as:

$$\mathrm{a}_{ij} = (\Phi^T\Phi)^{-1}\Phi^T\mathrm{y}_{ij} \tag{4}$$

where $\mathrm{y}_{ij} = \ln\big(X_{\text{norm}}(i,j,:)\big)$is the log-transformed decay profile, and a_{ij} contains the polynomial coefficients. A polynomial degree $n = 5$ is used to generate six coefficient images A_0 to A_5, consistent with TSR practice for CFRP [16]. Each coefficient image is independently min–max normalized to 8-bit.

2.4 Cyclic Alignment

Since the number of raw frames T varies between 383 and 2000 across the six specimens, whereas the number of processed images per channel is $m = 6$, a modular indexing approach is used, defining:

$$c_i = \big((i-1) \bmod m\big) + 1, i = 1,2,\dots,T \tag{5}$$

Each three-channel frame is then constructed by concatenating the i-th raw frame with the c_i-th PCT and TSR images:

$$\mathrm{I}_{3ch}(i) = F_{\text{norm}}(\cdot,\cdot,i) \;\|\; \mathcal{R}(P_{c_i}) \;\|\; \mathcal{R}(A_{c_i}) \tag{6}$$

where $\mathcal{R}(\cdot)$ denotes bilinear resampling to the 512×512 target resolution and $\|$ denotes concatenation along the channel dimension. The resulting tensor $\mathrm{I}_{3ch} \in \mathbb{R}^{T\times H\times W\times3}$ is analogous to an RGB video stream and does not require modification of pre-trained architectures.

The complete construction procedure is described in Algorithm 1 below.

Algorithm 1: Three-channel thermal image construction

1. **Input:** $X \in \mathbb{R}^{H\times W\times T\times1}$(raw thermal tensor); frame rate f_s [Hz]; polynomial degree $n = 5$; number of preprocessed images $m = 6$
2. **Step 1 (Global Normalisation):** Compute $X_{norm} = X/\max_{i,j,t}(X_{i,j,t})$
3. **Step 2 (PCT):** Reshape X_{norm} to $B \in \mathbb{R}^{T\times HW}$; subtract temporal mean; apply economy SVD; reshape first m right singular vectors to $H \times W$; min-max normalise each P_k to 8-bit
4. **Step 3 (TSR):** Build $\tau_k = \ln(k/f_s)$ for $k = 1, \dots, T$; form $\Phi \in \mathbb{R}^{T\times(n+1)}$; for each pixel (i, j), compute $\mathbf{a}_{ij} = (\Phi^T\Phi)^{-1}\Phi^T\mathbf{y}_{ij}$; min-max normalise each coefficient image A_p to 8-bit
5. **Step 4 (Frame Normalization):** Apply Equation 2 to each of the T raw frames
6. **Step 5 (Cyclic Stacking):** For $i = 1$ to T: compute $c_i = ((i-1) \bmod m) + 1$; concatenate $F_{norm}(\cdot,\cdot,i)$, $\mathcal{R}\{P_{c_i}\}$, $\mathcal{R}\{A_{c_i}\}$ along the channel axis
7. **Output:** $\boldsymbol{I}_{3ch} \in \mathbb{R}^{T\times H\times W\times3}$

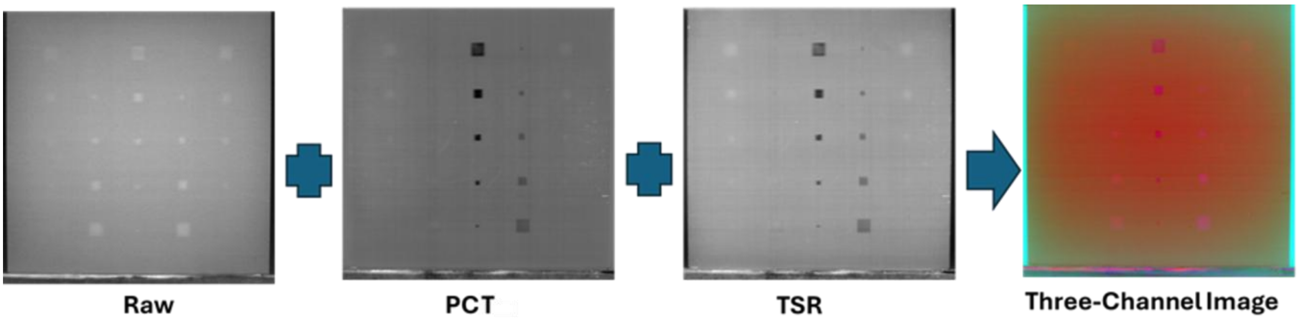


Figure 2. Example composite images for specimen SQ_01: Channel 1 (raw frame), Channel 2 (PC2 component), Channel 3 (A1 coefficient image)

3. Experimental Setup

3.1 Thermographic Experimental Datasets

The experimental thermal images used in this study are taken from six CFRP specimens belonging to two independent experimental datasets and are used for validation of the proposed framework. Both datasets are based on optical pulsed thermography (OPT), which is one of the most widely used non-contact non-destructive techniques for subsurface defect detection in composite materials. It enables thermal excitation and infrared imaging to be performed on the same surface of the specimen in a single-sided configuration. This setup is known as reflection mode, and its representation is shown in Figure 3 [36].

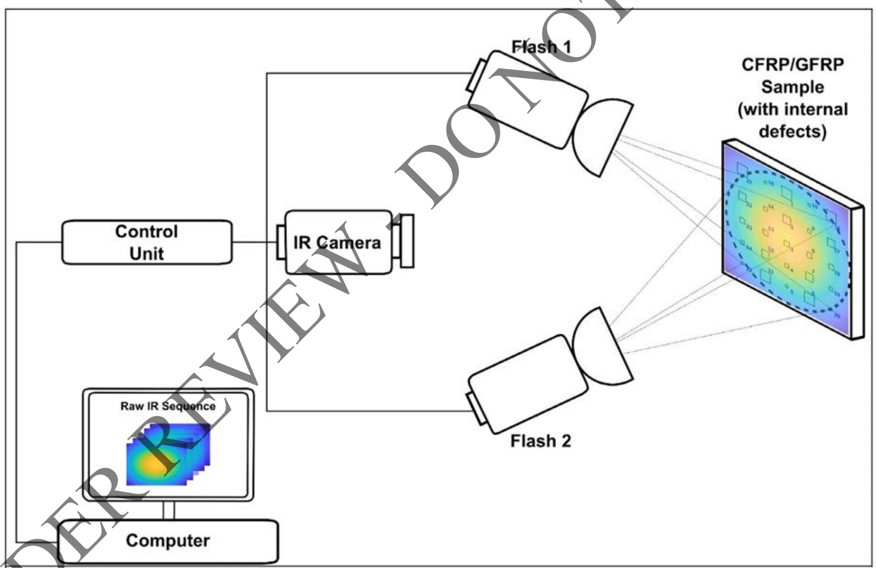
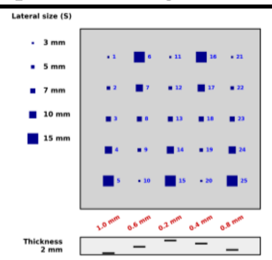
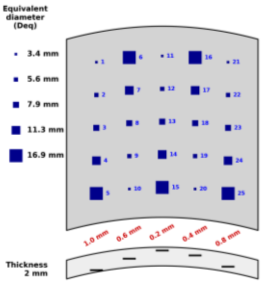
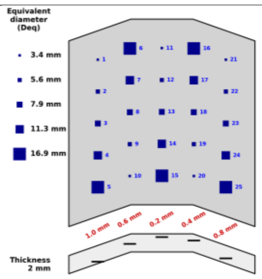


Figure 3. Schematic diagram of the OPT acquisition principle [36]

3.2 Dataset 1: Square-Insert CFRP Specimens

Dataset 1 is from Erazo-Aux et al. [36] and was acquired using a cooled InSb infrared camera (FLIR X6900sc, 512×512 pixels, 14-bit radiometric depth) at 120 to 145 Hz. The camera was positioned 100 cm from the specimen surface with energy supplied by two BALCAR FX60 flashes (12.8 kJ, 2 ms duration). Each acquisition captures approximately 2000 frames across a 16–17 s cooling window. The three specimens labelled SQ_01 (flat panel), SQ_02 (curved panel), and SQ_03 (trapezoidal panel) which correspond to specimens CFRP006, CFRP007 and CFRP008 in the original publication, consist of 300×300×2 mm quasi-isotropic CFRP laminates containing 25 square Teflon inserts in a 5×5 grid at nominal depths of 0.2, 0.4, 0.6, 0.8, and 1.0 mm. The geometrical variability between the three panels allows explicit testing of the representation's robustness against surface heating non-uniformity. Specimen characteristics are summarized in Table 1.

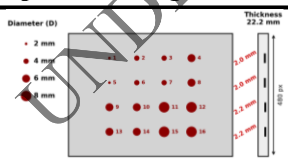
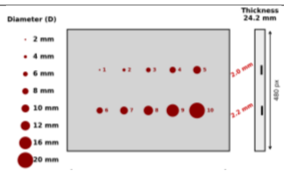
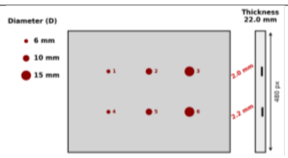
Table 1: Dataset 1 specimen characteristics

ID	Specimen Image	Geometry	Dimensions (mm)	Sizes (mm)	Depth (mm)	Frames
SQ_01		Flat	300 x 300 x 2	3, 6, 9, 12, 15	0.2–1.0	2000
SQ_02		Curved	300 x 300 x 2	3, 6, 9, 12, 15	0.2–1.0	2000
SQ_03		Trapezoidal	300 x 300 x 2	3, 6, 9, 12, 15	0.2–1.0	2000

3.3 Dataset 2: Circular-Insert CFRP Specimens

Dataset 2 is from Ahmed et al. [37] and uses an uncooled vanadium oxide microbolometer (FLIR A655sc, 640×480 pixels) at 50 Hz under halogen lamp illumination. Sequence lengths are 729, 465, and 383 frames for specimens labelled CI_01, CI_02, and CI_03 which correspond to specimens 3, 1 and 2 in the original publication respectively. These specimens contain circular Teflon inserts at depths of 2.0 mm and 2.2 mm, ranging from 2 mm to 20 mm in diameter. The shallow 0.2 mm separation between the two depth levels is a challenging imaging condition. Specimen CI_01 is the most difficult, containing 2 mm diameter inserts at depths of 2.0–2.2 mm, near the spatial resolution limit of the camera. Specimen characteristics are summarized in Table 2.

Table 2: Dataset 2 specimen characteristics

ID	Specimen Image	Geometry	Dimensions (mm)	Diameters (mm)	Depth (mm)	Frames
CI_01		Flat	250 x 250 x 22.2	2, 4, 6, 8	2.0–2.2	729
CI_02		Flat	250 x 250 x 24.2	2–20	2.0–2.2	465
CI_03		Flat	450 x 300 x 22.0	6, 10, 15	2.0–2.2	383

3.4 Baseline Input Preparation

For the raw single-channel baseline, each normalized thermal frame (Channel 1 only) was converted to a three-channel pseudo-color image using the parula colormap, so that all networks receive inputs of identical $H \times W \times T \times 3$ spatial format. Both the raw baseline and the proposed three-channel fusion inputs were trained using identical architectures, cross-validation splits, augmentation schemes, and hyperparameters. Any observed performance difference therefore arises solely from the input representation.

3.5 Dataset Partitioning and Cross-Validation

To avoid data leakage, splitting is performed at the specimen level rather than the frame level. Frame-level splitting introduces two problems: consecutive frames within a thermographic sequence are highly correlated, so random splits allow near-identical frames in both training and test sets; and since each training frame is augmented five times, a post-augmentation frame-level split would inflate test accuracy by allowing augmented copies of test frames into the training set. Specimen-level splitting resolves both issues.

Three-fold cross-validation is used, assigning one square-insert specimen and one circular-insert specimen per fold to maintain balanced geometrical representation. Specimen assignments and training set sizes are shown in Table 3.

All augmentation is applied only to the training set. Each training image and its ground-truth binary mask are augmented to produce five additional samples by horizontal flipping, vertical flipping, random rotation in $\mathcal{U}(-25°, +25°)$, and random scaling in $\mathcal{U}(0.75, 1.10)$. Images use bilinear interpolation; masks use nearest-neighbour interpolation to preserve defect boundary integrity.

Table 3: Specimen assignments and training frame counts for the three-fold cross-validation.

Fold	Training Specimens Raw (SQ + CI)/ aug×5)	Validation Specimens Raw (SQ + CI)	Test Specimens Raw (SQ + CI)
1	SQ_01, CI_01 (2000 + 729) x 5 = 13645	SQ_02, CI_02 (2000+465) = 2465	SQ_03, CI_03 (2000 + 383) = 2383
2	SQ_03, CI_03 (2000 + 383) x 5 = 11915	SQ_01, CI_01 (2000 + 729) = 2729	SQ_02, CI_02 (2000+465) = 2465
3	SQ_02, CI_02 (2000+465) x 5 = 12325	SQ_03, CI_03 (2000 + 383) = 2383	SQ_01, CI_01 (2000 + 729) = 2729

Ground-truth binary masks are generated using LabelMe, with P_2 or P_3 images used as annotation references because they present clearer defect boundaries than raw thermograms. One mask is generated per specimen, giving six masks in total. Bounding boxes are extracted as axis-aligned rectangles enclosing connected components of the binary masks. Data are exported into COCO JSON format for Faster R-CNN and RT-DETR-L, and into YOLO text format for YOLOv8n.

3.6 Detection Architectures and Training

Three detection architectures are evaluated. YOLOv8n [24] uses CSPDarknet as a backbone and a bidirectional FPN as a neck, with an anchor-free detection head that regresses bounding boxes at every spatial location. RT-DETR-L [26] uses a hybrid conv-transformer encoder with IoU-aware query selection and Hungarian bipartite matching, eliminating the need for non-maximum suppression. Faster R-CNN [22] uses a ResNet-50 FPN backbone with a region proposal network (RPN) in the first stage and parallel classification and regression heads on RoI-aligned features in the second stage. The detection model architectures are shown in Figure. 4.

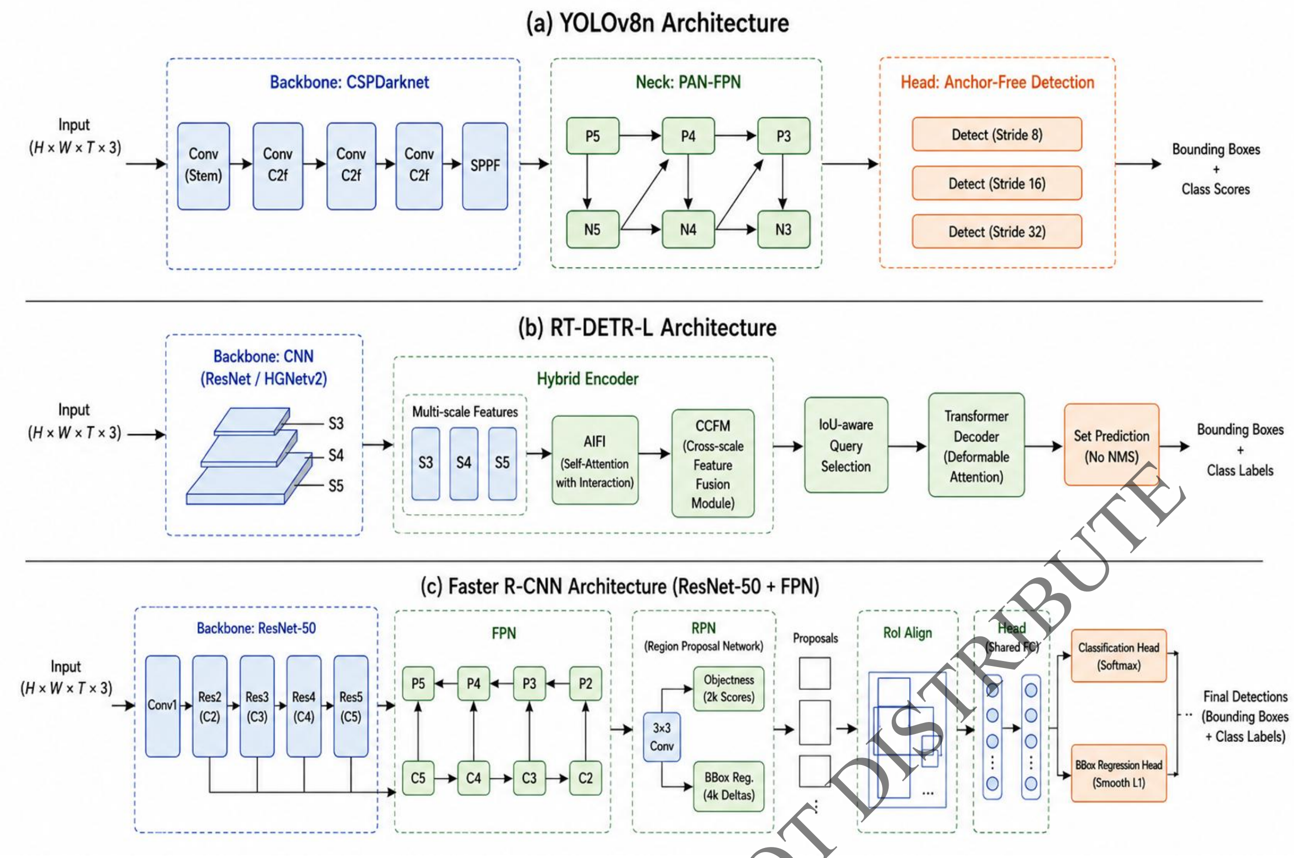


Figure 4. Detection model architectures: (a) YOLOv8n (CSPDarknet, anchor-free); (b) RT-DETR-L (hybrid conv–transformer, set prediction); (c) Faster R-CNN (ResNet-50 FPN, two-stage)

YOLOv8n and RT-DETR-L are trained using AdamW [38] with initial learning rate $\eta_0 = 0.001$, weight decay 5×10^{-4}, and a batch size of 16 with cosine annealing:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_0 - \eta_{min})\left(1 + \cos\left(\frac{\pi t}{T_{max}}\right)\right) \tag{7}$$

where t is the current epoch, $\eta_{\text{min}} = 1 \times 10^{-5}$, and $T_{\text{max}} = 50$. For RT-DETR-L, the learning rate of transformer attention layers is reduced to 1×10^{-4} to prevent divergence during early training. The YOLOv8n detection loss is:

$$\mathcal{L}_{det} = \mathcal{L}_{focal} + \lambda_{box}\,\mathcal{L}_{CIoU} \tag{8}$$

where $\mathcal{L}_{focal}$ is the focal classification loss, $\mathcal{L}_{CIoU}$ is the Complete IoU loss accounting for center distance, overlap, and aspect ratio, and $\lambda_{box} = 7.5$. The RT-DETR-L matching loss is:

$$\mathcal{L}_{DETR} = \sum_i \left(\mathcal{L}_{cls}(p_{\sigma(i)}, c_i) + \mathcal{L}_{box}(b_{\sigma(i)}, b_i)\right) \tag{9}$$

where σ is the permutation produced by the Hungarian matching algorithm [26], $p_{\sigma(i)}$ and $b_{\sigma(i)}$ are the predicted class probability and bounding box, and c_i, b_i are the ground-truth class label and bounding box.

Faster R-CNN [22] is trained using SGD with momentum 0.9, weight decay 1×10^{-4}, an initial learning rate of 2.5×10^{-4}, with a 1000-iteration linear warmup and a maximum of 3000 iterations using Detectron2. Anchor sizes of $\{32, 64, 128, 256, 512\}$ with aspect ratios of $\{0.5, 1, 2\}$ are used, and RoI Align with a 7×7 resolution is applied. All models use 576×704 pixel input with letterbox padding and early stopping (patience = 10 epochs). Hyperparameters are summarized in Table 4.

Table 4: Training hyperparameters for the three detection models.

Parameter	YOLOv8n	RT-DETR-L	Faster R-CNN
Parameters (M)	3.2	32	41
Optimiser	AdamW	AdamW	SGD (mom=0.9)
η_0	0.001	0.001 / 0.0001	0.00025
Weight decay	0.0005	0.0005	0.0001
Batch size	16	16	2
Max epochs / iterations	50 ep	50 ep	3000 it
LR schedule	Cosine	Cosine	Step (warmup)
Early stopping	10 ep	10 ep	-
Input resolution	576x704	576x704	576x704

3.7 Segmentation Architectures and Training

Three segmentation architectures are evaluated. U-Net [27] uses a ResNet-34 encoder [39] and connects encoder–decoder feature maps through concatenation-based skip connections. FPN [28] also uses a ResNet-34 encoder but connects feature pyramid levels in a top-down manner using element-wise summation. DeepLabV3+ [29] uses a ResNet-50 encoder with atrous spatial pyramid pooling at dilation rates $r \in \{1,6,12,18\}$, aggregating multi-scale features without reducing encoder resolution. The segmentation model architectures are shown in Figure 5.

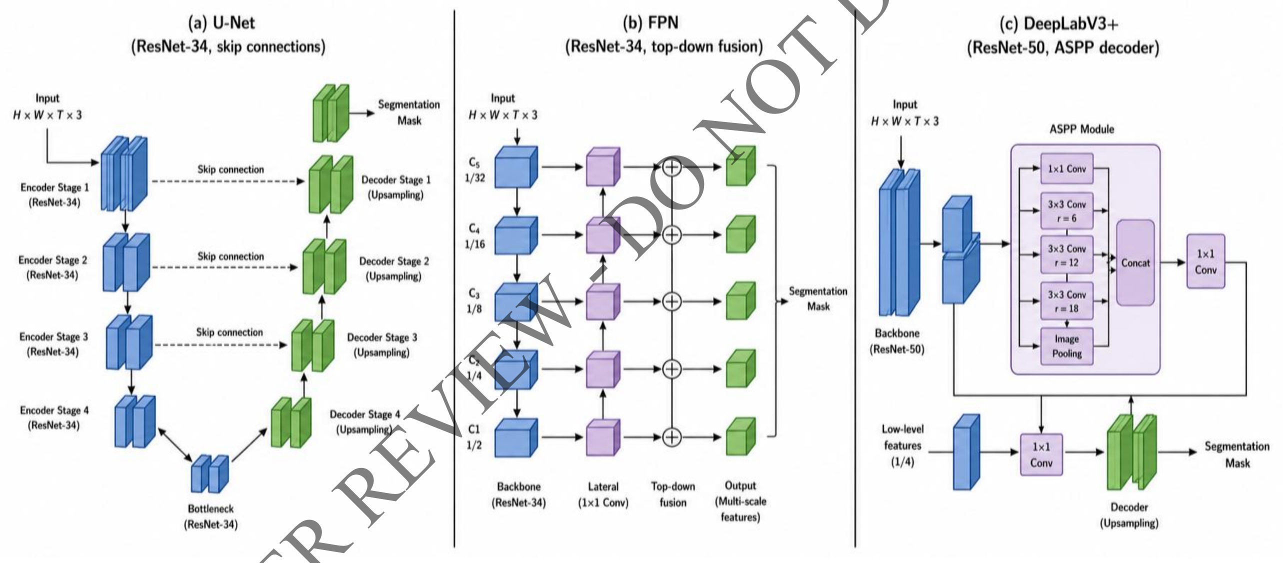


Figure 5. Segmentation model architectures: (a) U-Net (ResNet-34, skip connections); (b) FPN (ResNet-34, top-down fusion); (c) DeepLabV3+ (ResNet-50, ASPP decoder)

All three segmentation architectures are trained using AdamW [38] with an initial learning rate $\eta_0 = 1 \times 10^{-4}$, weight decay 5×10^{-4}, batch size of 8, and a cosine annealing (Eq. 7). Models use early stopping (patience = 10 epochs) and gradient clipping at ℓ_2-norm = 1.0. The segmentation loss combines binary cross-entropy and soft Dice loss:

$$\mathcal{L}_{seg} = \frac{1}{2}\mathcal{L}_{BCE} + \frac{1}{2}\mathcal{L}_{Dice} \quad (10)$$

The binary cross-entropy term includes label smoothing with $\varepsilon = 0.1$:

$$\mathcal{L}_{BCE} = -\frac{1}{N_p}\sum_{p=1}^{N_p}[\tilde{y}_p \ln(\hat{y}_p) + (1 - \tilde{y}_p)\ln(1 - \hat{y}_p)] \quad (11)$$

where N_p is the number of pixels within the batch, and $\tilde{y}_p = y_p(1 - \varepsilon) + \varepsilon/2$ is the smoothed ground-truth label. $\hat{y}_p$ is the predicted pixel probability. The soft Dice loss is:

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{p=1}^{N_p} y_p \hat{y}_p + \delta}{\sum_{p=1}^{N_p} y_p + \sum_{p=1}^{N_p} \hat{y}_p + \delta} \tag{12}$$

where $\delta = 10^{-6}$ is a smoothing constant. Full hyperparameter configurations are in Table 5.

Table 5: Training hyperparameters for the three segmentation models.

Parameter	U-Net	FPN	DeepLabV3+
Encoder	ResNet-34	ResNet-34	ResNet-50
Parameters (M)	21	28	41
Pretrained encoder	ImageNet	ImageNet	ImageNet
Optimiser	AdamW	AdamW	AdamW
$\boldsymbol{\eta_0}$	0.0001	0.0001	0.0001
Weight decay	0.0005	0.0005	0.0005
Batch size	8	8	8
Max epochs	40	40	40
LR schedule	Cosine	Cosine	Cosine
Early stopping	10 ep	10 ep	10 ep
Gradient clip norm	1.0	1.0	1.0
Label smoothing $\boldsymbol{\epsilon}$	0.1	0.1	0.1
ASPP dilation rates	-	-	1, 6, 12, 18

3.8 Evaluation Metrics

Detection models are evaluated using precision, recall, F1-score, and mAP@0.5, where a prediction is correct if the IoU between predicted and ground-truth bounding boxes exceeds 0.5:

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN} \tag{13}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

Segmentation models are evaluated using pixel-level IoU and Dice coefficient:

$$\text{Dice} = \frac{2\,TP}{2\,TP + FP + FN}, \ \ \text{IoU}_{px} = \frac{TP}{TP + FP + FN} \tag{15}$$

where TP are correctly predicted defect pixels, FP are incorrectly predicted background pixels, and FN are missed defect pixels. All experiments are performed on an NVIDIA Tesla P100 GPU (16 GB VRAM) using PyTorch 2.0.1, CUDA 11.8, and cuDNN 8.6.

4. Results and Discussion

To evaluate the effect of the proposed three-channel input representation, each architecture is trained separately under raw single-channel and three-channel fused input conditions while all other experimental parameters are held constant. Results are reported across all three cross-validation folds. For each fold, quantitative results are presented first, followed by visualizations and physical interpretation.

4.1 Detection Results

4.1.1 Performance on Fold 1

In Fold 1, models are trained on SQ_01 and CI_01 , validated on SQ_02 and CI_02, and tested on SQ_03 (trapezoidal plate with 25 square inserts, depths 0.2 mm to 1.0 mm) and CI_03 (flat plate with 6 circular inserts, diameters 6 mm, 10 mm, and 15 mm, depths 2.0 mm to 2.2 mm).

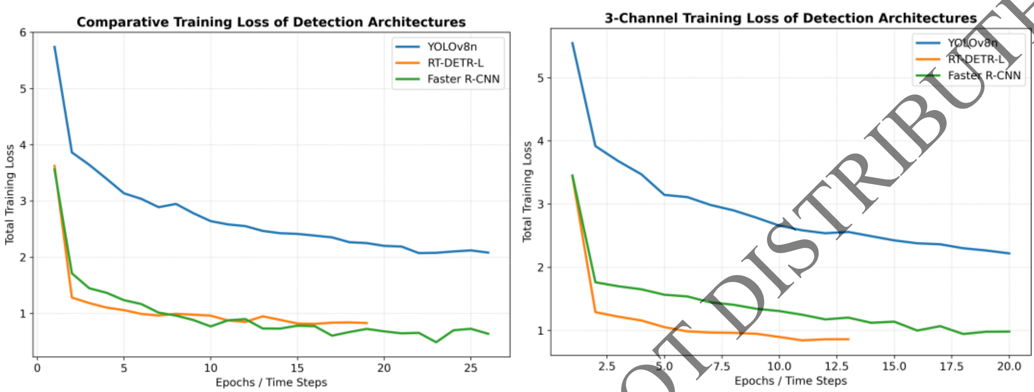


Figure 6. Detection training curves, single-channel vs. three-channel input, all three architectures, Fold 1

Figure 6 demonstrate training convergence for Fold 1. Comparing loss under raw single-channel input and three-channel input shows validation loss with raw single-channel input has more oscillation and more time for convergence (especially with circular defects). Validation loss converges earlier with three-channel input and loss converges at a more stable rate.

Table 6: Detection results for Fold 1 test specimens SQ_03 and CI_03.

Model	Specimen	Raw				3 Channel			
		Precision	Recall	F1 Score	mAP@0.5	Precision	Recall	F1 Score	mAP@0.5
YOLOv8n	SQ_03	0.95	0.90	0.93	0.94	0.96	0.94	0.95	0.96
	CI_03	0.88	0.75	0.81	0.82	0.95	0.87	0.91	0.91
	Mean	**0.92**	**0.86**	**0.89**	**0.89**	**0.96**	**0.91**	**0.93**	**0.94**
RT-DETR-L	SQ_03	0.96	0.92	0.94	0.95	0.98	0.95	0.96	0.97
	CI_03	0.91	0.80	0.85	0.85	0.96	0.86	0.91	0.92
	Mean	**0.94**	**0.88**	**0.91**	**0.91**	**0.97**	**0.91**	**0.94**	**0.95**
Faster R-CNN	SQ_03	0.94	0.84	0.89	0.88	0.95	0.88	0.91	0.91
	CI_03	0.86	0.68	0.76	0.75	0.90	0.81	0.85	0.86
	Mean	**0.91**	**0.77**	**0.83**	**0.84**	**0.93**	**0.85**	**0.88**	**0.89**

On SQ_03, all architectures achieve high mAP scores (0.88–0.95) under single-channel input, indicating adequate thermal contrast for square inserts at shallow depths. The three-channel input provides slight improvements across all models by producing more tightly constrained bounding boxes, where raw thermal halos blur defect boundaries. A more pronounced improvement is observed on CI_03, where Faster R-CNN improves from mAP = 0.75 to 0.86, a 14.7 percentage-point increase. The overall mean mAP improvement across both test specimens is 5.6% (YOLOv8n), 4.4% (RT-DETR-L), and 5.9% (Faster R-CNN).

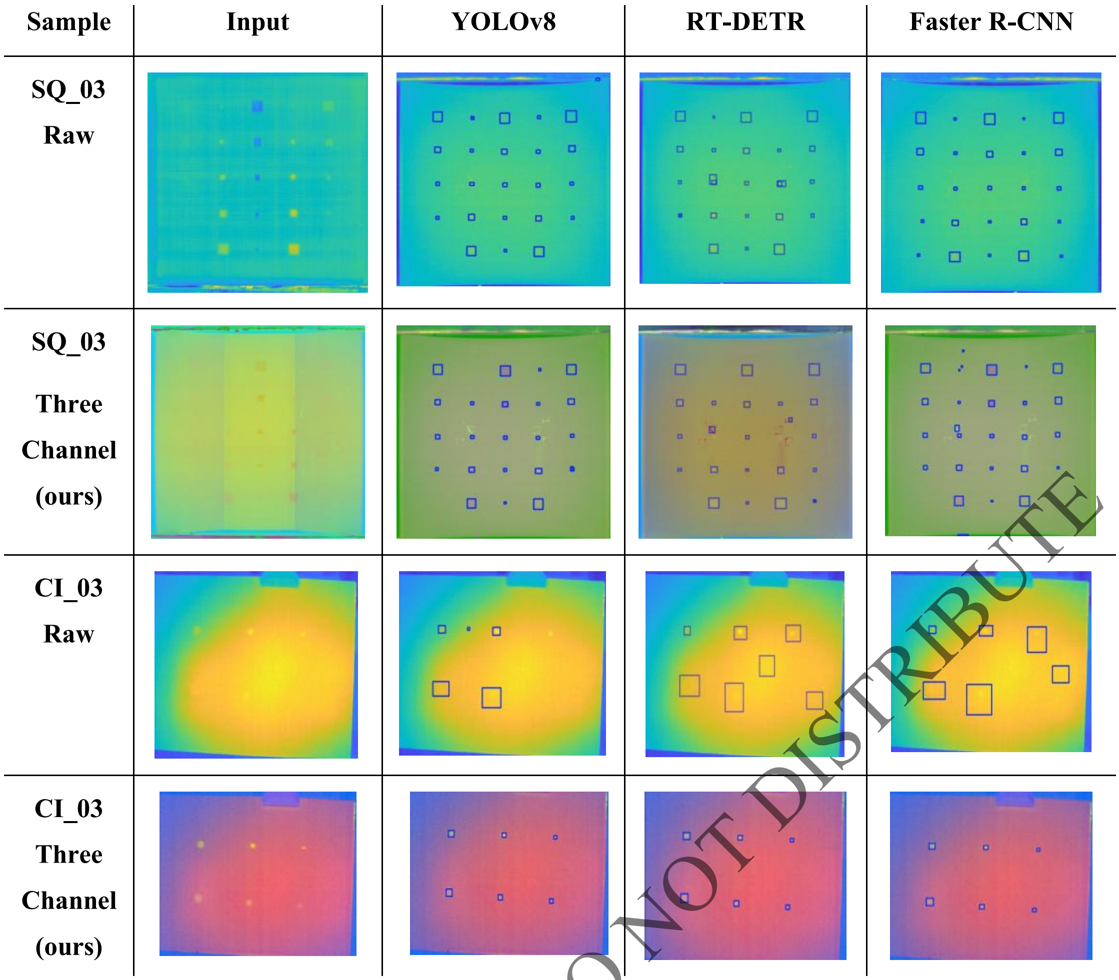


Figure 7. Detection visualizations for Fold 1 test specimens SQ_03 and CI_03.

The improvement on CI_03 is attributed to the reduction of diagonal illumination gradients captured by the PCT channel, which mask shallow circular defects at plate edges in raw representations, while the TSR channel restores temporal decay information that is partially lost due to frame-wise averaging.

4.1.2 Performance on Fold 2

In Fold 2, models are trained on SQ_03 and CI_03, validated on SQ_01 and CI_01, and tested on SQ_02 (curved plate with 25 square inserts) and CI_02 (flat plate with 10 circular inserts, diameters 2 mm to 20 mm, depths 2.0 mm to 2.2 mm). CI_02 represents the most diverse circular defect conFigureuration in this study and provides a strong test of generalization.

Table 7: Detection results for Fold 2 test specimens SQ_02 and CI_02.

Model	Specimen	Raw				3 Channel			
		Precision	Recall	F1 Score	mAP@0.5	Precision	Recall	F1 Score	mAP@0.5
YOLOv8n	SQ_02	0.93	0.60	0.73	0.60	0.94	0.68	0.79	0.68
	CI_02	0.83	0.50	0.62	0.50	0.85	0.60	0.70	0.60
	Mean	**0.88**	**0.55**	**0.68**	**0.55**	**0.90**	**0.64**	**0.75**	**0.65**
RT-DETR-L	SQ_02	0.95	0.80	0.86	0.80	0.57	0.80	0.66	0.80
	CI_02	0.66	0.60	0.63	0.70	0.83	1.00	0.90	1.00
	Mean	**0.81**	**0.70**	**0.75**	**0.75**	**0.70**	**0.90**	**0.78**	**0.90**
Faster R-CNN	SQ_02	0.94	0.64	0.76	0.64	0.94	0.72	0.81	0.72
	CI_02	0.00	0.00	0.00	0.00	0.81	0.90	0.69	0.90
	Mean	**0.47**	**0.32**	**0.38**	**0.32**	**0.88**	**0.81**	**0.75**	**0.81**

Fold 2 shows the largest improvement across all folds. Under single-channel input, RT-DETR-L achieves mAP = 0.70 on CI_02 and Faster R-CNN completely fails (mAP = 0.00). With the three-channel input, RT-DETR-L achieves mAP 1.00 and Faster R-CNN achieves 0.90, representing the largest single-specimen recovery in this study. YOLOv8n improves from mAP 0.50 to 0.60. The overall mean mAP improvement across both test specimens is 0.10 (YOLOv8n), 0.15 (RT-DETR-L), and 0.49 (Faster R-CNN).

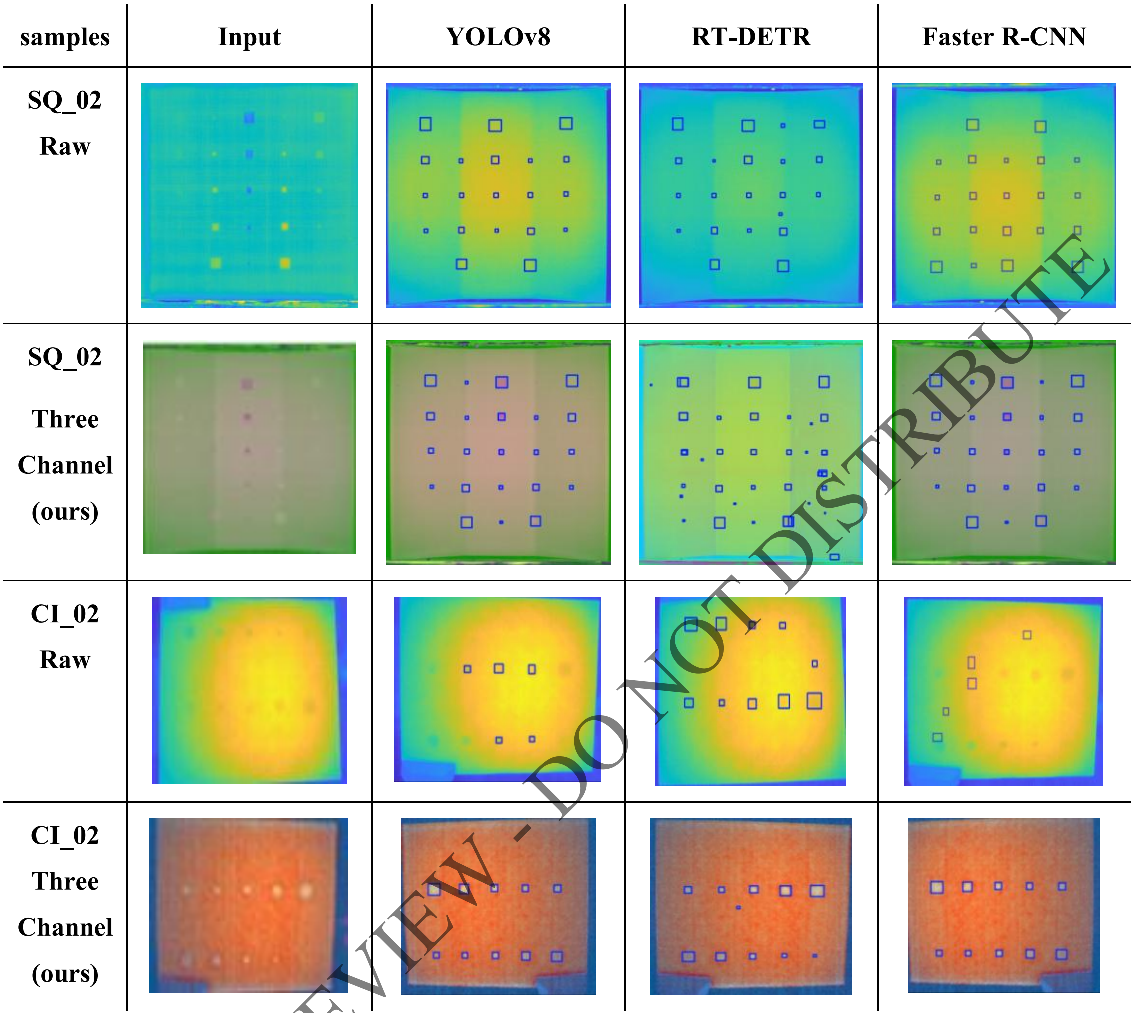


Figure 8. Detection visualizations for Fold 2 test specimens SQ_02 and CI_02.

The near-complete failure of Faster R-CNN on CI_02 under single-channel input is attributed to its two-stage architecture. The RPN filters candidate regions based on local intensity anomalies; when circular defects produce weak thermal contrast, such anomalies are absent and candidate regions are discarded before reaching the second-stage classifier. The three-channel representation mitigates this by introducing complementary spatial and temporal features. The strong performance of RT-DETR-L is attributed to its global attention mechanism, which effectively exploits the additional PCT and TSR information.

4.1.3 Performance on Fold 3

In Fold 3, models are trained on SQ_02 and CI_02, validated on SQ_03 and CI_03, and tested on SQ_01 (flat plate with 25 square inserts, depths 0.2 mm to 1.0 mm) and CI_01 (flat plate with 16 circular inserts, diameters 2 mm to 8 mm, depths 2.0 mm to 2.2 mm). The 2 mm diameter defects in CI_01 are the smallest in this study and represent the most challenging detection case, as their thermal footprint approaches the spatial resolution limit of the imaging system.

Table 8: Detection results for Fold 3 test specimens SQ_01 and CI_01.

Model	Specimen	Raw				3 Channel			
		Precision	Recall	F1 Score	mAP@0.5	Precision	Recall	F1 Score	mAP@0.5
YOLOv8n	SQ_01	0.94	0.68	0.77	0.79	0.95	0.76	0.84	0.76
	CI_01	0.25	0.06	0.10	0.06	0.83	0.31	0.45	0.31
	Mean	**0.60**	**0.37**	**0.44**	**0.43**	**0.89**	**0.54**	**0.65**	**0.54**
RT-DETR-L	SQ_01	0.82	0.76	0.79	0.76	0.95	0.80	0.86	0.76
	CI_01	0.00	0.00	0.00	0.00	0.92	0.75	0.82	0.75
	Mean	**0.41**	**0.38**	**0.40**	**0.38**	**0.94**	**0.78**	**0.84**	**0.76**
Faster R-CNN	SQ_01	0.90	0.72	0.80	0.72	0.94	0.72	0.81	0.72
	CI_01	0.00	0.00	0.00	0.00	0.36	0.25	0.29	0.25
	Mean	**0.45**	**0.36**	**0.40**	**0.36**	**0.65**	**0.49**	**0.55**	**0.49**

Both RT-DETR-L and Faster R-CNN produce zero detections on CI_01 under single-channel input, as does YOLOv8n near-completely (mAP = 0.06). With the three-channel representation, RT-DETR-L achieves mAP = 0.75, YOLOv8n achieves 0.31, and Faster R-CNN achieves 0.25. However, none of the architectures achieve reliable detection of 2 mm diameter defects under either representation. With a 640×480 camera, 2 mm defects occupy very few pixels, resulting in insufficient thermal signal for reliable detection. This imposes a fundamental lower bound on detectable defect size that cannot be overcome by preprocessing or representation design alone.

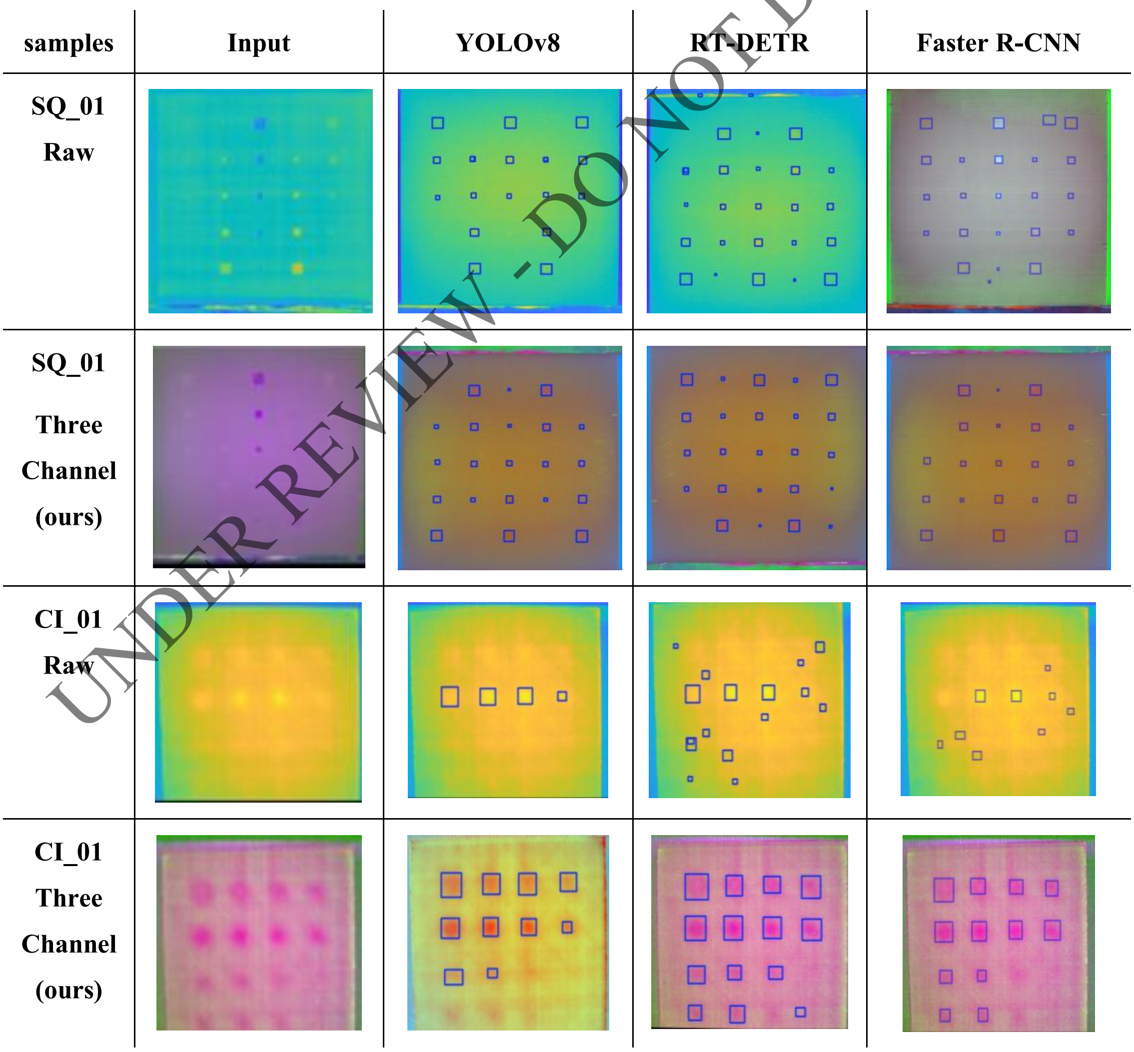


Figure 9. Detection visualizations for Fold 3 test specimens SQ_01 and CI_01.

4.2 Segmentation Results

4.2.1 Performance on Fold 1

The segmentation models in Fold 1 are trained on SQ_01 and CI_01, validated on SQ_02 and CI_02, and tested on SQ_03 and CI_03, using the same specimen geometry as discussed for Fold 1 detection performance above.

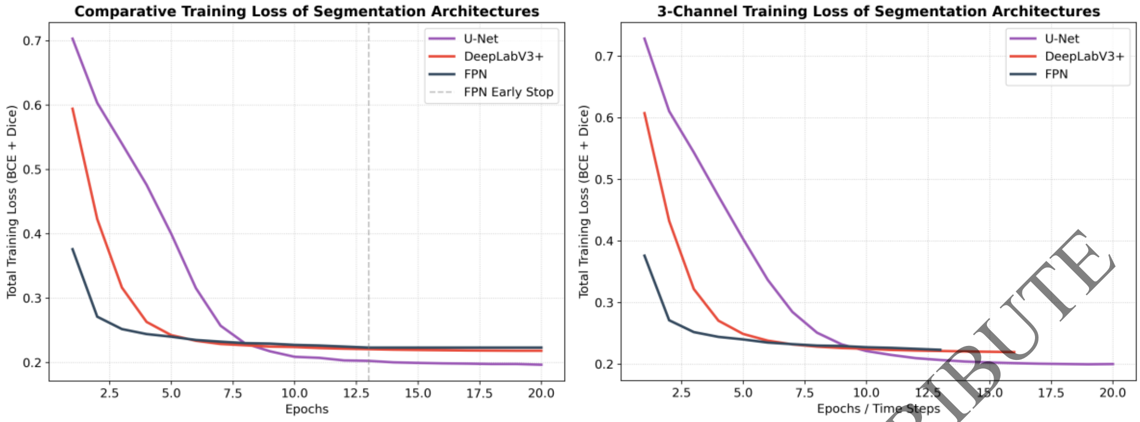


Figure 10. Segmentation training curves, single-channel vs. three-channel input, all three architectures, Fold 1

Figure 10. illustrates training convergence for Fold 1. Single-channel inputs produce higher loss oscillation and slower convergence, especially on circular specimens. FPN and DeepLabV3+ show poor loss reduction on circular specimens under raw input. All three models converge faster and more smoothly with the three-channel input, with early stopping occurring between epochs 22 and 31, indicating that the input representation limits convergence speed.

Table 9: Segmentation results, Fold 1. Test specimens SQ_03 and CI_03.

Model	Specimen	Raw				3 Channel			
		Precision	Recall	Dice	IoU	Precision	Recall	Dice	IoU
U-Net	SQ_03	0.92	0.76	0.83	0.72	0.92	0.75	0.84	0.72
	CI_03	0.19	0.59	0.29	0.17	0.19	0.96	0.34	0.20
	Mean	**0.56**	**0.68**	**0.56**	**0.44**	**0.56**	**0.86**	**0.59**	**0.46**
FPN	SQ_03	0.89	0.85	0.87	0.77	0.89	0.75	0.85	0.73
	CI_03	0.14	0.96	0.25	0.14	0.14	0.97	0.39	0.24
	Mean	**0.52**	**0.90**	**0.56**	**0.46**	**0.52**	**0.86**	**0.62**	**0.49**
DeepLabV3+	SQ_03	0.92	0.75	0.83	0.71	0.92	0.76	0.83	0.72
	CI_03	0.14	0.76	0.23	0.13	0.14	0.75	0.27	0.16
	Mean	**0.53**	**0.75**	**0.53**	**0.42**	**0.53**	**0.76**	**0.55**	**0.44**

All three models perform similarly well on SQ_03 with Dice scores between 0.83 and 0.87 under both input formats, indicating that raw single-channel input is sufficient for high-contrast square targets and that the three-channel format does not degrade performance on such specimens. On CI_03, U-Net recall increases from 0.59 to 0.96, indicating that the network captures nearly all defect pixels with the three-channel input. FPN shows high recall under both inputs but improves Dice from 0.25 to 0.39 by tightening boundary predictions. DeepLabV3+ shows a modest improvement.

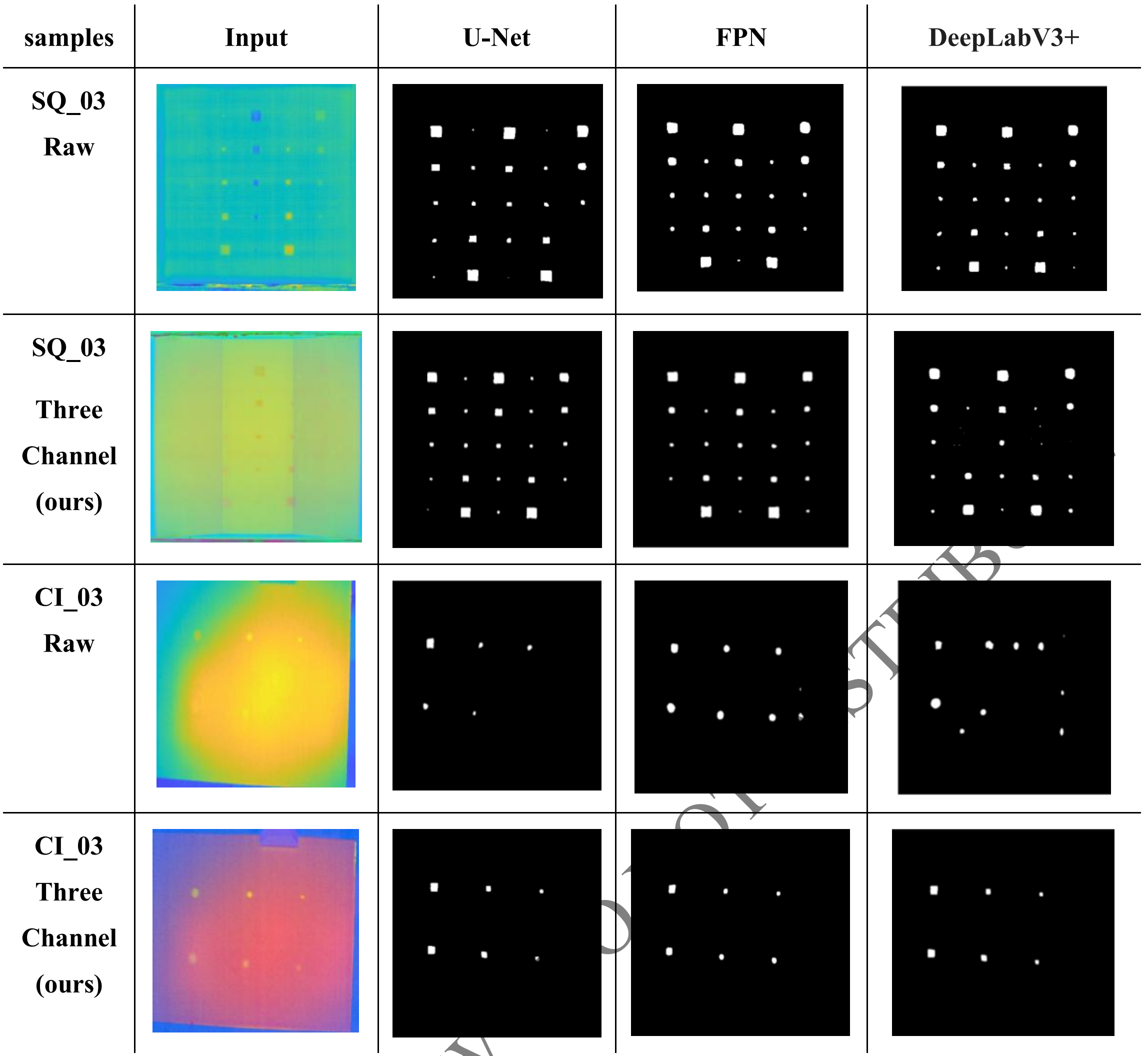


Figure 11. Segmentation mask visualizations for Fold 1 on SQ_03 and CI_03.

4.2.2 Performance on Fold 2

The segmentation models in Fold 2 are trained on SQ_03 and CI_03, validated on SQ_01 and CI_01, and tested on SQ_02 and CI_02.

Table 10: Segmentation results, Fold 2. Test specimens SQ_02 and CI_02.

Model	Specimen	Raw				3 Channel			
		Precision	Recall	Dice	IoU	Precision	Recall	Dice	IoU
U-Net	SQ_02	0.79	0.86	0.82	0.70	0.95	0.75	0.84	0.72
	CI_02	0.50	0.50	0.50	0.33	0.42	0.69	0.52	0.35
	Mean	**0.64**	**0.68**	**0.66**	**0.51**	**0.68**	**0.72**	**0.68**	**0.53**
FPN	SQ_02	0.87	0.83	0.85	0.74	0.86	0.85	0.85	0.75
	CI_02	0.00	0.00	0.00	0.00	0.67	0.20	0.31	0.18
	Mean	**0.43**	**0.41**	**0.42**	**0.37**	**0.76**	**0.52**	**0.58**	**0.46**
DeepLabV3+	SQ_02	0.83	0.89	0.86	0.76	0.84	0.88	0.86	0.76
	CI_02	0.19	0.04	0.06	0.03	0.55	0.37	0.44	0.28
	Mean	**0.51**	**0.46**	**0.46**	**0.39**	**0.69**	**0.62**	**0.65**	**0.52**

Fold 2 shows the most substantial segmentation improvements. FPN fails completely on CI_02 under raw input at Dice = 0.00, recovering to Dice 0.31 with the three-channel input. DeepLabV3+ improves from Dice = 0.06 to 0.44. U-Net shows marginal Dice improvement 0.66 to 0.68, though precision on SQ_02 increases from 0.79 to 0.95. FPN's sensitivity to raw single-channel input is attributed to its element-wise addition in the top-down pathway, which attenuates weak spatial variations, whereas the concatenation-based skip connections in U-Net partially compensate for this. The additional contrast provided by the TSR channel improves feature separability for deep circular defects in the intermediate feature maps.

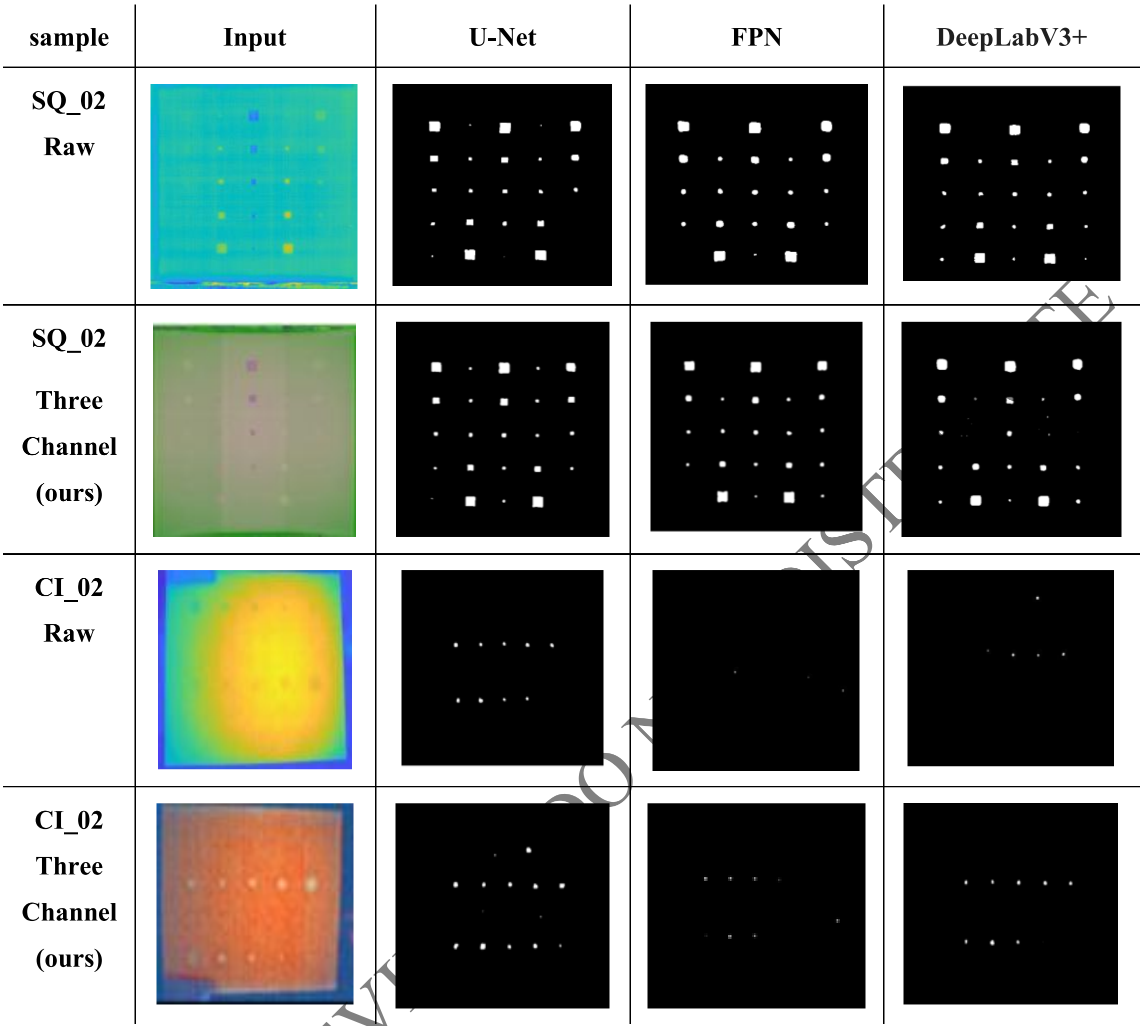


Figure 12. Segmentation mask visualizations for Fold 2 on SQ_02 and CI_02.

4.2.3 Performance on Fold 3

The segmentation models in Fold 3 are trained on SQ_02 and CI_02, validated on SQ_03 and CI_03, and tested on SQ_01 and CI_01.

Table 11: Segmentation results, Fold 3. Test specimens SQ_01 and CI_01.

Model	Specimen	Raw				3 Channel			
		Precision	Recall	Dice	IoU	Precision	Recall	Dice	IoU
U-Net	SQ_01	0.85	0.84	085	0.74	0.86	0.86	0.86	0.75
	CI_01	0.49	0.08	0.15	0.08	0.47	0.19	0.26	0.15
	Mean	**0.67**	**0.46**	**0.050**	**0.41**	**0.66**	**0.52**	**0.56**	**0.45**
FPN	SQ_01	0.82	0.86	0.84	0.72	0.85	0.84	0.85	0.74
	CI_01	0.36	0.07	0.11	0.06	0.53	0.11	0.19	0.10
	Mean	**0.59**	**0.46**	**0.47**	**0.39**	**0.69**	**0.47**	**0.52**	**0.42**
DeepLabV3+	SQ_01	0.82	0.87	0.85	0.73	0.84	0.84	0.84	0.73
	CI_01	0.21	0.08	0.11	0.06	0.43	0.13	0.20	0.11
	Mean	**0.51**	**0.47**	**0.48**	**0.39**	**0.63**	**0.48**	**0.52**	**0.42**

All three models exhibit low Dice scores on CI_01 under raw input (0.11–0.15) with very low recall (0.07–0.08), indicating fewer than two of the sixteen circular defects are correctly segmented on average. With the three-channel input, recall increases to 0.11–0.19 and Dice improves to 0.19–0.26, corresponding to approximately 3–4 detectable defects. U-Net achieves a 73% relative Dice improvement; FPN and DeepLabV3+ achieve 73% and 82%, respectively. Despite these gains, overall Dice scores remain low. As observed in the detection experiments for this fold, the 2 mm circular defects at 2.0–2.2 mm depth are near the physical resolution limit of the imaging system and cannot be fully resolved regardless of input representation.

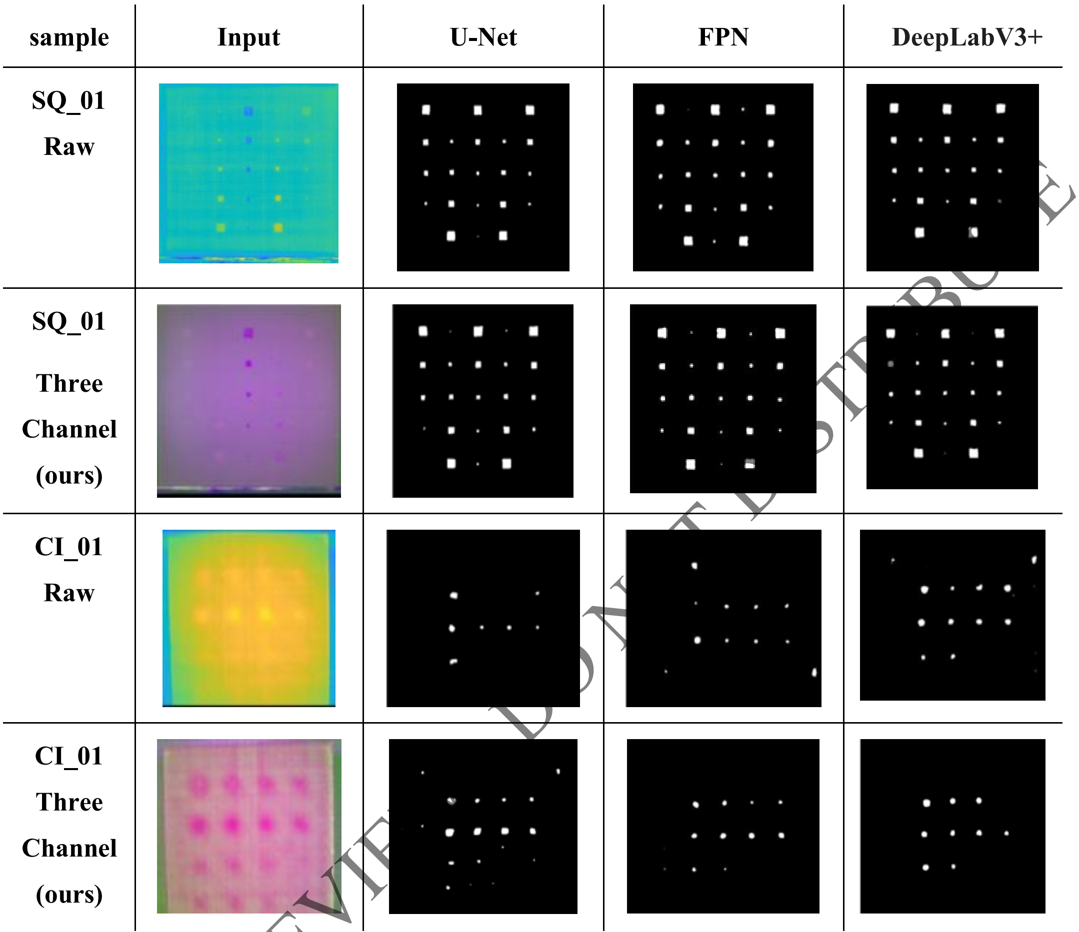


Figure 13. Segmentation mask visualizations for Fold 2 on SQ_01 and CI_01

4.3 Cross-Architecture Analysis

The combined results across all six models and three folds show a consistent trend. Regardless of segmentation architecture, Dice scores for circular specimens under single-channel input remain within a narrow range (0.11–0.29), with an overall mean of 0.24. This confirms that no architecture provides a meaningful advantage under raw input, and that the primary performance bottleneck lies in the input representation rather than the model design. The three-channel representation increases the cross-architecture, cross-fold average Dice to 0.33, with average improvements of 14% for DeepLabV3+, 16% for U-Net, and 51% for FPN. For detection, the cross-architecture mean mAP increases from 0.47 to 0.68, with Faster R-CNN showing the most pronounced improvement due to increased proposal sensitivity. Performance on square specimens remains largely unchanged across both tasks, confirming that the proposed representation selectively enhances learning for low-contrast circular defects rather than globally altering model behavior.

5. Conclusion

In this paper we proposed a cyclic-aligned three-channel thermographic fusion representation for deep learning-based detection and segmentation of subsurface defects in CFRP using pulsed thermography. The representation combines raw thermal intensity, PCT spatial components, and TSR polynomial coefficients into a $T \times H \times W \times 3$ tensor, addressing the mismatch between variable-length thermographic sequences and fixed network input channels, while remaining compatible with standard pre-trained architectures without any architectural modification. Experiments were conducted using three-fold cross-validation across six models (YOLOv8n, RT-DETR-L, Faster R-CNN, U-Net, FPN, DeepLabV3+) on two independent datasets, yielding consistent improvements of 4–11% in mAP for detection and substantially improved segmentation Dice scores for deep circular defects, while maintaining performance on simpler defect geometries.

The consistent gains across architectures confirm that the input representation is the primary limiting factor in thermographic defect analysis pipelines. The PCT channel suppresses illumination non-uniformities and enhances defect visibility; the TSR channel encodes depth-dependent thermal decay. Together they provide complementary physical information unavailable from single-channel input. A key limitation is the lack of evaluation on real industrial defects with more diverse geometries and acquisition conditions. Future work should extend the evaluation to such settings, investigate adaptation of the representation to different thermographic setups, and explore integration of the preprocessing steps into an end-to-end trainable framework.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

CRediT Authorship Contribution Statement

Zain Ul Abidin: Writing – original draft, Conceptualization, Data curation, Methodology, Software, Formal analysis, Investigation
Habeeban Memon: Data curation, Writing – review & editing, Validation, Software.
Junaid Ahmed: Supervision, Project administration, Conceptualization, Validation, Writing – review & editing.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the authors used generative AI models to refine the academic language and improve structural flow. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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References

[1] Mouritz, A.P. "Introduction to Aerospace Materials". *Woodhead Publishing*, Sawston, 2012. https://doi.org/10.2514/4.869198

[2] L. Mishnaevsky, K. Branner, H. N. Petersen, J. Beauson, M. McGugan, and B. F. Sørensen, "Materials for wind turbine blades: An overview," *Materials*, vol. 10, no. 11, p. 1285, 2017. https://doi.org/10.3390/ma10111285

[3] R. H. Bossi and V. Giurgiutiu, "Nondestructive testing of damage in aerospace composites," in *Polymer Composites in the Aerospace Industry*, pp. 413–448, 2015. https://doi.org/10.1016/B978-0-85709-523-7.00015-3

[4] S. Abrate, *Impact on Composite Structures*. Cambridge, UK: Cambridge University Press, 1998. https://doi.org/10.1017/CBO9780511574504

[5] F. Ciampa, M. Pooya, P. Fulvio, and M. Michele, "Recent advances in active infrared thermography for non-destructive testing of aerospace components," *Sensors*, vol. 18, no. 2, p. 609, 2018. https://doi.org/10.3390/s18020609

[6] I. Papa, V. Lopresto, and A. Langella, "Ultrasonic inspection of composites materials: Application to detect impact damage," *International Journal of Lightweight Materials and Manufacture*, vol. 4, no. 1, pp. 37–42, 2021. https://doi.org/10.1016/j.ijlmm.2020.04.002

[7] P. J. Schilling, B. R. Karedla, A. K. Tatiparthi, M. A. Verges, and P. D. Herrington, "X-ray computed microtomography of internal damage in fiber reinforced polymer matrix composites," *Composites Science and Technology*, vol. 65, no. 14, pp. 2071–2078, 2005. https://doi.org/10.1016/j.compscitech.2005.05.014

[8] X. P. Maldague, *Theory and Practice of Infrared Technology for Nondestructive Testing*. New York, NY, USA: Wiley, 2001

[9] P. Cawley and R. D. Adams, "The mechanics of the coin-tap method of non-destructive testing," *Journal of Sound and Vibration*, vol. 122, no. 2, pp. 299–316, 1988. https://doi.org/10.1016/s0022-460x(88)80356-0

[10] C. Meola and G. M. Carlomagno, "Recent advances in the use of infrared thermography," *Measurement Science and Technology*, vol. 15, no. 9, pp. R27–R58, 2004. https://doi.org/10.1088/0957-0233/15/9/R01

[11] R. Yang and Y. He, "Optically and non-optically excited thermography for composites: A review," *Infrared Physics & Technology*, vol. 75, pp. 26–50, 2016. https://doi.org/10.1016/j.infrared.2015.12.026

[12] A. Moskovchenko, M. Švantner, V. Vavilov, and A. Chulkov, "Analyzing probability of detection as a function of defect size and depth in pulsed IR thermography," *NDT & E International*, vol. 130, p. 102673, 2022. https://doi.org/10.1016/j.ndteint.2022.102673

[13] P. Zhu, R. Wang, S. Sfarra, V. Vavilov, X. Maldague, and H. Zhang, "A novel force-temperature model for evaluating internal forces in CFRP by means of infrared thermography," *NDT & E International*, vol. 143, p. 103066, 2024. https://doi.org/10.1016/j.ndteint.2024.103066

[14] G. Poelman, S. Hedayatrasa, J. Segers, W. Van Paepegem, and M. Kersemans, "Adaptive spectral band integration in flash thermography: Enhanced defect detectability and quantification in composites," *Composites Part B: Engineering*, vol. 202, p. 108305, 2020. https://doi.org/10.1016/j.compositesb.2020.108305

[15] N. Rajic, "Principal component thermography for flaw contrast enhancement and flaw depth characterisation in composite structures," *Composite Structures*, vol. 58, no. 4, pp. 521–528, 2002. https://doi.org/10.1016/S0263-8223(02)00161-7

[16] S. M. Shepard, J. R. Lhota, B. A. Rubadeux, D. Wang, and T. Ahmed, "Reconstruction and enhancement of active thermographic image sequences," *Optical Engineering*, vol. 42, no. 5, pp. 1337–1342, 2003. https://doi.org/10.1117/1.1566969

[17] X. Maldague and S. Marinetti, "Pulse phase infrared thermography," *Journal of Applied Physics*, vol. 79, no. 5, pp. 2694–2698, 1996. https://doi.org/10.1063/1.362662

[18] Y. He, B. Deng, H. Wang, L. Cheng, K. Zhou, S. Cai, and F. Ciampa, "Infrared machine vision and infrared thermography with deep learning: a review," *Infrared Physics & Technology*, vol. 116, p. 103754, 2021. https://doi.org/10.1016/j.infrared.2021.103754
[19] R. G. Rosa, B. P. Barella, I. G. Vargas, J. R. Tarpani, H.-G. Herrmann, and H. Fernandes, "Advanced thermal imaging processing and deep learning integration for enhanced defect detection in carbon fiber-reinforced polymer laminates," *Materials*, vol. 18, no. 7, p. 1448, 2025. https://doi.org/10.3390/ma18071448
[20] D. G. Lema, O. D. Pedrayes, R. Usamentiaga, P. Venegas, and D. F. García, "Automated Detection of Subsurface Defects Using Active Thermography and Deep Learning Object Detectors," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–13, 2022. https://doi.org/10.1109/TIM.2022.3169484
[21] Z. Tong, L. Cheng, S. Xie, and M. Kersemans, "A flexible deep learning framework for thermographic inspection of composites," *NDT & E International*, vol. 139, p. 102926, 2023. https://doi.org/10.1016/j.ndteint.2023.102926
[22] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017. https://doi.org/10.1109/TPAMI.2016.2577031
[23] Q. Fang, C. Ibarra-Castanedo, I. Garrido, Y. Duan, and X. Maldague, "Automatic detection and identification of defects by deep learning algorithms from pulsed thermography data," *Sensors*, vol. 23, no. 9, p. 4444, 2023. https://doi.org/10.3390/s23094444
[24] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," version 8.0.0, 2023. [Online]. Available: https://github.com/ultralytics/ultralytics
[25] S. Schmid, J. Reinhardt, and C. U. Grosse, "Spatial and temporal deep learning for defect detection with lock-in thermography," *NDT & E International*, vol. 143, p. 103063, 2024. https://doi.org/10.1016/j.ndteint.2024.103063
[26] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16965–16974, 2024. https://doi.org/10.1109/CVPR52733.2024.01605
[27] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Munich, Germany, 2015, pp. 234–241. https://doi.org/10.1007/978-3-319-24574-4_28
[28] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 2017, pp. 2117–2125. https://doi.org/10.1109/CVPR.2017.106
[29] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 801–818, 2018. https://doi.org/10.1007/978-3-030-01234-2_49
[30] D. Morelli, R. Marani, E. D'Accardi, D. Palumbo, U. Galietti, and T. D'Orazio, "A Convolution Residual Network for Heating-Invariant Defect Segmentation in Composite Materials Inspected by Lock-in Thermography," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–14, 2021. https://doi.org/10.1109/TIM.2021.3116300
[31] M. Salah, N. Werghi, D. Svetinovic, and Y. Abdulrahman, "Multi-modal attention networks for enhanced segmentation and depth estimation of subsurface defects in pulse thermography," *arXiv preprint arXiv:2501.09994*, 2025. https://arxiv.org/abs/2501.09994
[32] Y. Liu, X. Zhang, J. Wang, and Y. Zhao, "Automatic defect detection and three-dimensional reconstruction from pulsed thermography images based on a bidirectional long-short term memory network," *Engineering Applications of Artificial Intelligence*, vol. 124, p. 106574, 2023. https://doi.org/10.1016/j.engappai.2023.106574

[33] J. Xie, Z. Zeng, Y. Ma, Y. Pan, X. Wu, X. Han, and Y. Tian, "Defect recognition in sonic infrared imaging by deep learning of spatiotemporal signals," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108174, 2024. https://doi.org/10.1016/j.engappai.2024.108174
[34] G. Zhou, Z. Zhang, W. Yin, H. Chen, L. Wang, D. Wang, and H. Ma, "Surface defect detection of CFRP materials based on infrared thermography and attention U-Net algorithm," *Nondestructive Testing and Evaluation*, vol. 39, no. 2, pp. 238–257, 2024. https://doi.org/10.1080/10589759.2023.2191954
[35] B. Hu, B. Gao, W. L. Woo, L. Ruan, J. Jin, Y. Yang, and Y. Yu, "A lightweight spatial and temporal multi-feature fusion network for defect detection," *IEEE Transactions on Image Processing*, vol. 30, pp. 472–486, 2021. https://doi.org/10.1109/TIP.2020.3036770
[36] J. Erazo-Aux, H. Loaiza-Correa, A. D. Restrepo-Girón, C. Ibarra-Castanedo, and X. Maldague, "Thermal imaging dataset from composite material academic samples inspected by pulsed thermography," *Data in Brief*, vol. 32, p. 106313, 2020. https://doi.org/10.1016/j.dib.2020.106313
[37] J. Ahmed, B. Gao, and W. L. Woo, "Wavelet-Integrated Alternating Sparse Dictionary Matrix Decomposition in Thermal Imaging CFRP Defect Detection," *IEEE Transactions on Industrial Informatics*, vol. 15, no. 7, pp. 4033–4043, 2019. https://doi.org/10.1109/TII.2018.2881341
[38] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proceedings of the International Conference on Learning Representations (ICLR)*, New Orleans, LA, USA, 2019. https://openreview.net/forum?id=Bkg6RiCqY7
[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016, pp. 770–778. https://doi.org/10.1109/CVPR.2016.90